CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ SOLUTIONS

Chapter 5: Quadratic Equations

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 The quadratic formula

1. Solve $2x^2 - 5x - 4 = 0$, giving the answers correct to three significant figures. ICSE 2025, 4 marks

SOLUTION

$a = 2, b = -5, c = -4$. $D = 25 + 32 = 57$ and $\sqrt{57} \approx 7.550$.

$x = \dfrac{5 \pm 7.550}{4}$, so $x = \dfrac{12.550}{4} = 3.1375$ or $x = \dfrac{-2.550}{4} = -0.6375$

Answer: $x = 3.14$ **or** $x = -0.637$ (using $\sqrt{57} = 7.5498$, the second root is -0.63746)

2. Solve $2x^2 - 10x + 5 = 0$, giving the answers correct to three significant figures. ICSE 2024, 3 marks

SOLUTION

$a = 2, b = -10, c = 5$. $D = 100 - 40 = 60$ and $\sqrt{60} \approx 7.746$.

$x = \dfrac{10 \pm 7.746}{4}$, so $x = \dfrac{17.746}{4} = 4.4365$ or $x = \dfrac{2.254}{4} = 0.5635$

Answer: $x = 4.44$ **or** $x = 0.564$

3. Solve $x^2 + 4x - 8 = 0$, giving the answers correct to one decimal place. ICSE 2023, 4 marks

SOLUTION

$a = 1, b = 4, c = -8$. $D = 16 + 32 = 48$ and $\sqrt{48} \approx 6.928$.

$x = \dfrac{-4 \pm 6.928}{2}$, so $x = 1.464$ or $x = -5.464$

Answer: $x = 1.5$ **or** $x = -5.5$

4. Solve $x^2 - 7x + 3 = 0$, giving the answers correct to two decimal places. ICSE 2020, 3 marks

SOLUTION

$a = 1, b = -7, c = 3$. $D = 49 - 12 = 37$ and $\sqrt{37} \approx 6.083$.

$x = \dfrac{7 \pm 6.083}{2}$, so $x = 6.5415$ or $x = 0.4585$

Answer: $x = 6.54$ **or** $x = 0.46$

5. Solve $x^2 - 4x - 8 = 0$, giving the answers correct to three significant figures. ICSE 2019, 3 marks

SOLUTION

$D = 16 + 32 = 48$ and $\sqrt{48} \approx 6.928$

$x = \dfrac{4 \pm 6.928}{2}$, so $x = 5.464$ or $x = -1.464$

Answer: $x = 5.46$ **or** $x = -1.46$

6. Solve $x^2 + 7x = 7$, giving the answers correct to two decimal places. ICSE 2018, 4 marks

SOLUTION

Write it as $x^2 + 7x - 7 = 0$. $D = 49 + 28 = 77$ and $\sqrt{77} \approx 8.775$.

$x = \dfrac{-7 \pm 8.775}{2}$, so $x = 0.8875$ or $x = -7.8875$

Answer: $x = 0.89$ **or** $x = -7.89$

7. Solve $4x^2 - 5x - 3 = 0$, giving the answers correct to two decimal places. ICSE 2017, 4 marks

SOLUTION

$D = 25 + 48 = 73$ and $\sqrt{73} \approx 8.544$

$x = \dfrac{5 \pm 8.544}{8}$, so $x = 1.693$ or $x = -0.443$

Answer: $x = 1.69$ **or** $x = -0.44$

2 Nature of the roots

8. The equation $3x^2 + \sqrt{7}x + 2 = 0$ has: (a) two equal real roots (b) two distinct real roots (c) more than two real roots (d) no real roots ICSE 2025, 1 mark

SOLUTION

$D = b^2 - 4ac = 7 - 24 = -17$, which is negative.

Answer: (d)

9. The roots of $px^2 - qx + r = 0$ are real and equal when: (a) $p^2 = 4qr$ (b) $q^2 = 4pr$ (c) $-q^2 = 4pr$ (d) $p^2 > 4qr$ ICSE 2024, 1 mark

SOLUTION

Equal roots need $D = 0$: $(-q)^2 - 4pr = 0$, so $q^2 = 4pr$.

Answer: (b)

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10. Find the values of k for which $x^2 + 4kx + (k^2 - k + 2) = 0$ has equal roots. ICSE 2018, 3 marks

SOLUTION

Equal roots need $D = 0$: $(4k)^2 - 4(k^2 - k + 2) = 0$

$12k^2 + 4k - 8 = 0$, that is $3k^2 + k - 2 = 0$

$(3k - 2)(k + 1) = 0$

Answer: $k = \frac{2}{3}$ **or** $k = -1$

3 Standard form and roots

11. 3 is a root of $x^2 - px + 3 = 0$. Find p. (a) 4 (b) 3 (c) 5 (d) 2 ICSE 2023, 1 mark

SOLUTION

Put $x = 3$: $9 - 3p + 3 = 0$, so $p = 4$.

Answer: (a)

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